# Package 'rProject'

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**Title** Setup R Project

**Version** 0.0.59

**Description** A collection of functions to setup R projects.

**URL** https://github.com/ijapesigan/rProject/, https://ijapesigan.github.io/rProject/

**BugReports** https://github.com/ijapesigan/rProject/issues

**License** MIT + file LICENSE

**Encoding** UTF-8

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`AI` *Write the Project's AI Instructions*

---

## Description

Writes the instruction files that AI tools read in this project, on top of the global instructions installed by `AIGlobal()`.

## Usage

```
AI(path, type = NULL)
```

## Arguments

| | |
|---|---|
| `path` | Character string. Project path. |
| `type` | Character vector. Project types: any of `"package"`, `"cran"` (a package on CRAN or headed there; with `"package"`), `"manuscript"`, `"simulation"`, `"library"`, `"talk"`, and `"container"`. |

## Details

- `.setup/ai/project.md`: project facts that the user maintains. Created once and never overwritten.
- `.setup/todo/`: the user's to-do list, `todo.txt`, which agents read when planning and never edit, with a `.gitignore` that keeps the folder out of the public repository. Created only if the folder does not exist.
- `AGENTS.md` (Codex, Copilot CLI, Claude through `CLAUDE.md`), `CLAUDE.md` (Claude Code), and `.github/copilot-instructions.md` (Copilot in the IDE): generated, and rewritten when they change.

All of them are private (gitignored and excluded from the package build).

The project type comes from `type`, else from an `ai` list in `.setup/scripts/project.R` (e.g., `ai <- list(type = c("manuscript", "simulation"))`), else from the template type in `.setup/scripts/project.R` (`type <- "cran"`; see `Template()`), else it is detected: `"simulation"` if `.sim/README.md` (which describes the simulation) exists, `"manuscript"` if a `*-manuscript*.Rtex` exists or the name starts

with man, "package" and "cran" for a cran-* folder (its project.R names the package), "library" for lib-*, "talk" for talks*, "container" for docker-*, otherwise "package".

Where the project lives (the private GitHub repository, the Overleaf project, and the public GitHub repository) comes from .setup/repo/, which Project() creates when it is missing.

The user chooses the Apptainer image and runs apptainer exec; the instructions tell agents not to pick an image.

## Value

Returns the paths of the files that were written, invisibly.

## Author(s)

Ivan Jacob Agaloos Pesigan

---

AIGlobal *Install Global AI Instructions*

---

## Description

Copies the global AI instructions and Claude settings from .setup/ai/global in the project to the configuration folders of Claude Code, Codex, and GitHub Copilot in the home directory. Files are written only when their content changed. Existing files that differ are backed up with the suffix .bak.

## Usage

```
AIGlobal(path, home = Sys.getenv("HOME"))
```

## Arguments

| | |
|---|---|
| path | Character string. Project path. |
| home | Character string. Home directory. |

## Details

| Source | Installed to |
|---|---|
| AGENTS.md | ~/.claude/CLAUDE.md |
| AGENTS.md | ~/.codex/AGENTS.md |
| AGENTS.md | ~/.copilot/copilot-instructions.md |
| claude-settings.json | ~/.claude/settings.json |

claude-settings.json is merged into an existing settings.json: its keys win, permission rules are added to the existing ones, and other keys are kept.

### Value

Returns the paths of the files that were written.

### Author(s)

Ivan Jacob Agaloos Pesigan

---

## Bib *Consolidate .bib Files into a Single File*

---

### Description

Joins the `.bib` files in `.setup/latex/bib` into `bib.bib`, verbatim and in file-name order, and copies it to `.setup/quarto/bib/quarto.bib` and `vignettes/vignettes.bib` when those folders exist. Without sources (a clone of the repository, which has only `bib.bib`), the existing `bib.bib` is copied. The source files are never rewritten: in `lib-*` projects they are the entries themselves (one entry per file, ASCII with LaTeX accents). `LibCheck()` validates them with biber.

### Usage

```
Bib(path, bib_lib = FALSE)
```

### Arguments

| | |
|---|---|
| `path` | Character string. Project path. |
| `bib_lib` | Logical. If `bib_lib = TRUE`, build `bib.bib` from the public `lib-*` GitHub repositories (`make bib` does so in rProject, `make bibrproject` anywhere). If none can be cloned, the previous `bib.bib` is kept. |

### Author(s)

Ivan Jacob Agaloos Pesigan

---

## BibLib *Copy .bib Files from lib-* GitHUb Repositories*

---

### Description

Copy .bib Files from lib-* GitHUb Repositories

### Usage

```
BibLib(path)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

---

## Binary *Setup Binaries not Included in the Package*

---

### Description

Runs (sources) every R script in `.setup/bin/`, at `make project`; for example, a script that builds a bespoke binary the project needs. `.setup/bin/` is optional: `make` does not create it, and without it `Binary()` does nothing.

### Usage

```
Binary(path)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

---

## Build *Build Package*

---

### Description

Build Package

### Usage

```
Build(path, vignettes = FALSE, dependencies = FALSE)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| vignettes | Logical. Build vignettes. |
| dependencies | Logical. Install package dependencies. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## BuildIgnore *Build* .Rbuildignore

### Description

Build .Rbuildignore

### Usage

```
BuildIgnore(path, project, add = NULL)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| project | Character string. Project name. |
| add | Character vector. Items to add to .Rbuildignore. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## CFF *Build* CITATION.cff

### Description

Build CITATION.cff

### Usage

```
CFF(path)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## Clean *Clean Files Generated on Build*

### Description

Delete all output files that are generated on automated build.

### Usage

```
Clean(path, push = FALSE)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| push | Logical. If push = TRUE clean in preparation for git push: generated files are kept, and files matching *secrets* are deleted, except in hidden folders such as .setup. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## CleanPkg *Clean R Packages*

### Description

Remove the project's R library (.library). Packages outside the project (the image's site library, the user library) are never touched.

### Usage

```
CleanPkg(path)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## CleanProj *Clean Project*

### Description

Remove all files except `project.Rproj` and, optionally, `.library`.

### Usage

```
CleanProj(path, lib = FALSE, sim = FALSE, sif = FALSE)
```

### Arguments

| | |
|---|---|
| `path` | Character string. Project path. |
| `lib` | Logical. If `lib = TRUE`, delete `.library`. If `lib = FALSE`, `.library` is not deleted. |
| `sim` | Logical. If `sim = TRUE`, delete `.sim`. If `sim = FALSE`, `.sim` is not deleted. |
| `sif` | Logical. If `sif = TRUE`, delete `.sif`. If `sif = FALSE`, `.sif` is not deleted. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## CleanTinyTex *Clean ~/.TinyTex*

### Description

Uninstall `TinyTex`.

### Usage

```
CleanTinyTex(path)
```

### Arguments

| | |
|---|---|
| `path` | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## ConfigFiles *Setup Config Files*

### Description

Clones a public `dotfiles` GitHub repository and runs `make` in it. The clone ignores the global Git configuration, so URL rewrites to SSH host aliases do not apply and the repository is fetched anonymously over HTTPS. This also works inside a container without access to SSH keys.

### Usage

```
ConfigFiles(repo = "ijapesigan/dotfiles")
```

### Arguments

| | |
|---|---|
| `repo` | Character string. GitHub repository in the form `owner/name`. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## Coverage *Code Coverage*

### Description

Runs `covr::package_coverage()` on the package. To leave files out, put covr's `.covrignore` at the project root (and add it to `.Rbuildignore` with `ignore` in `.setup/scripts/project.R`).

### Usage

```
Coverage(path)
```

### Arguments

| | |
|---|---|
| `path` | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

`DataAnalysis` *Data Analysis*

## Description

Execute R scripts in `data-analysis`.

## Usage

```
DataAnalysis(path, clean = FALSE)
```

## Arguments

| | |
|---|---|
| `path` | Character string. Project path. |
| `clean` | Logical. Remove all `.Rds` files in `data-process` and `data-analysis` before running. |

## Author(s)

Ivan Jacob Agaloos Pesigan

`DataCompress` *Compress Data Files*

## Description

Compresses data files in place with xz (e.g., `data.txt` becomes `data.txt.xz`), checks that the compressed copy reads back byte for byte, and then removes the original. Compressed files are binary, so they do not count as editable text in Overleaf (see `SizeCheck()`). Scripts that find files with `DataPath()` keep working unchanged.

## Usage

```
DataCompress(path, files)
```

## Arguments

| | |
|---|---|
| `path` | Character string. Project path. |
| `files` | Character vector. Files to compress, relative to `path`. |

## Value

Returns the paths of the compressed files, invisibly.

## Author(s)

Ivan Jacob Agaloos Pesigan

## DataManifest *Write the Data Manifest*

### Description

Writes `.setup/ai/data.md` (private), with one row for each file in the data tiers (`.setup/data-raw`, `.setup/data-private`, `.setup/data-external`): the file, its tier, size, MD5, status, the source and location (filled in by the user and kept on rewrite), and the scripts that use it.

### Usage

```
DataManifest(path)
```

### Arguments

| | |
|---|---|
| `path` | Character string. Project path. |

### Details

External files are local only. One that is listed but missing on this machine is kept, marked "not on this machine"; one whose MD5 differs from the recorded one is marked "changed".

### Value

Returns the manifest as a data frame, invisibly.

### Author(s)

Ivan Jacob Agaloos Pesigan

## DataPath *Find a Data File in the Data Tiers*

### Description

Finds a data file by its plain name in the data tiers, in this order: `.setup/data-raw` (shareable), `.setup/data-private` (restricted; never public), and `.setup/data-external` (local only). The file can be stored as is or compressed (`.xz` or `.gz`); `utils::read.table()` and similar functions read compressed files directly.

### Usage

```
DataPath(path, file, plain = FALSE)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| file | Character string. Plain file name, e.g., "data.txt", even if it is stored as data.txt.xz. |
| plain | Logical. If TRUE, return a path to an uncompressed copy. |

### Details

With plain = TRUE, a compressed file is unpacked into .setup/data-cache (local only), for tools that need plain text, such as Mplus. The copy is refreshed when the compressed file changes.

### Value

Returns the path of the file.

### Author(s)

Ivan Jacob Agaloos Pesigan

---

## DataProcess *Process Data*

---

### Description

Execute R scripts in data-process.

### Usage

```
DataProcess(path, clean = FALSE)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| clean | Logical. Remove all .Rds files in data-process before running. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## Dependencies *Copy Dependencies to from* `.setup/r-dependencies` *to* `R`

### Description

Copy Dependencies to from `.setup/r-dependencies` to `R`

### Usage

```
Dependencies(path)
```

### Arguments

| | |
|---|---|
| `path` | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## Detritus *Create* `detritus/tex/tex-fig-*`

### Description

Create `detritus/tex/tex-fig-*`

### Usage

```
Detritus(path)
```

### Arguments

| | |
|---|---|
| `path` | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## Environment *Setup* .Renviron

### Description

Setup .Renviron

### Usage

```
Environment(path, project, tokens)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| project | Character string. Project name. |
| tokens | Named vector of tokens. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## EnvironmentProject *Set* PROJECT *Environment Variable*

### Description

Set PROJECT Environment Variable

### Usage

```
EnvironmentProject(project)
```

### Arguments

| | |
|---|---|
| project | Character string. Project name. |

### Value

Returns project name.

### Author(s)

Ivan Jacob Agaloos Pesigan

## EnvironmentTokens *Set Tokens*

### Description

Set Tokens

### Usage

```
EnvironmentTokens(tokens)
```

### Arguments

| | |
|---|---|
| tokens | Named vector of tokens. |

### Value

Returns vector of secret tokens.

### Author(s)

Ivan Jacob Agaloos Pesigan

## InstallLocal *Install Applications on Local Machine*

### Description

Install Applications on Local Machine

### Usage

```
InstallLocal(
  what = c("tmux", "julia", "mpdemo", "lazygit", "fastfetch", "quarto"),
  all = FALSE,
  force = FALSE
)
```

### Arguments

| | |
|---|---|
| what | Character string. Vector of applications to install. Valid values include "tmux", "julia", "mpdemo", "lazygit", "fastfetch", and "quarto". |
| all | Logical. If all = TRUE, ignore what and install all valid what values. |
| force | Logical. If force = TRUE, install whether or not the application is available in the system. |

### Author(s)

Ivan Jacob Agaloos Pesigan

---

`LatexCompile` *Compile Output of* `LatexKnit`

---

### Description

Runs `latexmk` on every `.tex` file, with 10 million extra words of pdfTeX main memory (`extra_mem_bot`), so that a long bibliography compiles. A document whose last run failed (fatal error, TeX capacity exceeded) stops `make latex` in GitHub Actions and gives a warning locally: `latexmk -f` would otherwise leave a stub PDF and succeed.

### Usage

```
LatexCompile(path, clean = FALSE)
```

### Arguments

| | |
|---|---|
| `path` | Character string. Project path. |
| `clean` | Logical. Clean `latexmk` output except `*.pdf` |

### Author(s)

Ivan Jacob Agaloos Pesigan

---

`LatexKnit` *Knit* `.setup/latex/*.Rtex`

---

### Description

Knit `.setup/latex/*.Rtex`

### Usage

```
LatexKnit(path)
```

### Arguments

| | |
|---|---|
| `path` | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## LatexMake *Compile Output of* LatexKnit

### Description

Compile Output of LatexKnit

### Usage

```
LatexMake(path, clean = FALSE)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| clean | Logical. Clean latexmk output except *.pdf |

### Author(s)

Ivan Jacob Agaloos Pesigan

## LatexTikzCompile *Compile Tikz*

### Description

Compile Tikz

### Usage

```
LatexTikzCompile(path, clean = FALSE)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| clean | Logical. Clean latexmk output except *.pdf |

### Author(s)

Ivan Jacob Agaloos Pesigan

`LibAdd` *Add an Entry to the lib-* Library from a DOI*

## Description

Drafts a biblatex entry from the DOI's Crossref record, in the house layout, and names the repository and file it belongs in:

## Usage

```
LibAdd(doi, root, key = NULL, title = NULL, fields = NULL, write = FALSE)
```

## Arguments

| | |
|---|---|
| `doi` | Character string. The DOI. |
| `root` | Character string. The folder with local clones of the `lib-*` projects. |
| `key` | Character string or `NULL`. A key to use instead of the drafted one (e.g., with a suffix). |
| `title` | Character string or `NULL`. A title to use instead of the drafted one. |
| `fields` | Named character vector or `NULL`. Fields to set or replace (e.g., full author names or an abstract from the publisher's page). |
| `write` | Logical. If `TRUE`, write the file and run `LibCheck()` on its repository. |

## Details

- the entry type and the repository type from the Crossref type;
- `date` from the print issue (`published-print`), else the journal issue, else the first publication; the key year and the decade repository follow it;
- names and text with LaTeX accents, and `\%` and `\&` escaped;
- the key as `Family-Family-Family[-etal]-Year`, accents dropped;
- the abstract without its JATS tags;
- the ISSN and publisher reused from the library's other entries for the same journal;
- the title in a sentence-case draft.

It never invents a field: what Crossref does not have is left out. It lists what it cannot decide (initials only, an online-first date, a missing abstract, the title's proper nouns, a key already in use) as questions for the user.

## Value

Returns a list with `file`, `text`, and `questions`, invisibly.

## Author(s)

Ivan Jacob Agaloos Pesigan

`LibCheck` *Check the Entries of a lib-* Library*

## Description

Checks every entry file in `.setup/latex/bib` of a `lib-<decade>-<type>` project against the house conventions.

## Usage

```
LibCheck(path, root = NULL, error = FALSE)
```

## Arguments

| | |
|---|---|
| `path` | Character string. Project path. |
| `root` | Character string or `NULL`. A folder with local clones of the other `lib-*` projects, to find keys and DOIs used in more than one repository. If `NULL`, only this project is checked. |
| `error` | Logical. If `TRUE`, stop when there is an ERROR. |

## Details

- **ERROR:** the file does not parse or holds more than one entry; the file name is not `<repo>-<key>.bib`; the key does not follow `Family-Family-Family[-etal]-Year[a-z]` (three names, then `etal`, for four or more authors); the key year differs from the `date` year or lies outside the decade; a required field is missing; a non-ASCII character; an unescaped `%` in a text field; a key or DOI used twice; biber reports a missing mandatory field.
- **WARNING:** an entry type outside the repository's type; a lone suffix (`b` without `a`); other biber datamodel warnings.
- **STYLE:** not in the house layout (see `LibFormat()`); `\a'`-style accents; a page range with one hyphen; a DOI with a prefix; no final newline.

biber's warning that `publisher` is not an `@Article` field is the house convention and is not reported.

## Value

Returns a data frame with the columns `file`, `key`, `severity`, `check`, and `message`, invisibly.

## Author(s)

Ivan Jacob Agaloos Pesigan

`LibFormat` *Put the Entries of a lib-* Library in the House Layout*

## Description

Rewrites each entry file in `.setup/latex/bib` of a `lib-*` project in the house layout, without changing what the entry says:

## Usage

```
LibFormat(path, write = FALSE)
```

## Arguments

| | |
|---|---|
| `path` | Character string. Project path. |
| `write` | Logical. If `TRUE`, rewrite the files that change. If `FALSE`, only report them. |

## Details

- `@Type{Key,`, then one `field = {value},` per line, then `}`, and a final newline;
- fields in the house order: `author`, `editor`, `bookauthor`, `booktitle`, `date`, `journaltitle`, `title`, `subtitle`, the other fields alphabetically, then `abstract`, `publisher`, `keywords`, `library`, and `annotation`;
- field names in lower case and entry types in the house spelling (`Article`, `InBook`, `PhdThesis`, ...);
- `\a'`-style accents as `\'` (`{\a'e}` becomes `{\'e}`);
- page ranges with `--`;
- DOIs without a `https://doi.org/` or `doi:` prefix.

Files that do not parse, or that hold more than one entry, are left as they are (see `LibCheck()`).

## Value

Returns the names of the files that change (or changed), invisibly.

## Author(s)

Ivan Jacob Agaloos Pesigan

## LibPaths *Add* .library *to Library Paths*

### Description

- creates a folder .library on the project path if it does not exist
- creates a .gitignore file inside .library if it does not exist
- adds .library to .libPaths()
- sets R_LIBS_USER to .library

### Usage

```
LibPaths(path)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |

### Value

Returns the path of .library.

### Author(s)

Ivan Jacob Agaloos Pesigan

## License *Generate License File*

### Description

Generate License File

### Usage

```
License(path, type = "mit")
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| type | Character string. License type. Available values include "mit" and "gpl3". |

### Author(s)

Ivan Jacob Agaloos Pesigan

## Lint *Lint Project*

### Description

Lint Project

### Usage

```
Lint(path)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## Lock *Record What Built the Project*

### Description

Writes `.setup/scripts/rproject.lock` at `make auto`, before the project is published. The lock records

- the rProject that is running (`rproject_sha`, `rproject_version`);
- the image (`image`, a registry digest, and `image_name`), read from the running sif's labels;
- the Posit Package Manager snapshot (`snapshot`, the image's date) for packages the image lacks, and the R version (`r_version`);
- the packages the project adds in `project.R` (`extra_cran`, `extra_github`), with their versions or commits;
- for a proprietary add-on sif (`<name>-plus.sif`, built from the public image with `Bootstrap: localimage`): the sif's name, source, checksum (`sif_sha256`), and date. CI runs the public image.

`make project` restores the lock, locally and in GitHub Actions; `make upgrade` moves rProject forward, and the next `make auto` records the running image.

### Usage

```
Lock(path)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |

### Details

docker-* projects have no lock: their recurring image builds take the newest upstream base and the newest rProject.

Nothing is written in rProject itself (its CI builds rProject from source), or when the running rProject has no commit (installed from a local source).

### Value

Returns the path of the lock file, invisibly, or NULL when nothing was written.

### Author(s)

Ivan Jacob Agaloos Pesigan

---

## Log *Create or Update a Change-Set Log*

---

### Description

Creates .setup/logs/YYYY-MM-DD-<topic>.md, the log for one change set (one Overleaf push), and adds a pending row to .setup/logs/index.md. The log has the commit message skeleton, a "Pushes" table, and the list of files to update in Overleaf.

### Usage

```
Log(
  path,
  topic,
  summary = topic,
  private = file.path(path, ".verify"),
  update = FALSE
)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| topic | Character string. Short topic in kebab case (e.g., "log-spec"). The change-set log uses the same topic. |
| summary | Character string. First line of the commit message. |
| private | Character string. Path to a clone of the private repository at the pushed commit. By default .verify in the project. |
| update | Logical. If TRUE, refresh only the file lists of the newest existing log for topic. |

### Details

The file lists come from comparing the project with a clone of the private repository (by default `.verify`, kept by `.setup/scripts/verify.sh`), using the same rules as `Verify()`. Without a clone, the lists hold `TODO`.

### Value

Returns the path of the log invisibly.

### Author(s)

Ivan Jacob Agaloos Pesigan

---

## `Manual` *Package Manual*

---

### Description

Package Manual

### Usage

```
Manual(path, project)
```

### Arguments

| | |
|---|---|
| `path` | Character string. Project path. |
| `project` | Character string. Project name. |

---

## `PkgDevel` *Install Development Packages from CRAN*

---

### Description

Install Development Packages from CRAN

### Usage

```
PkgDevel(path)
```

### Arguments

| | |
|---|---|
| `path` | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## PkgDevelGitHub *Install Development Packages from GitHub*

### Description

Install Development Packages from GitHub

### Usage

```
PkgDevelGitHub(path)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## PkgProject *Install Project Packages from CRAN*

### Description

Install Project Packages from CRAN

### Usage

```
PkgProject(path, pkg)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| pkg | Character vector. R packages. |

### Author(s)

Ivan Jacob Agaloos Pesigan

PkgProjectBioconductor *Install Project Packages from Bioconductor*

## Description

Install Project Packages from Bioconductor

## Usage

```
PkgProjectBioconductor(path, pkg)
```

## Arguments

| | |
|---|---|
| path | Character string. Project path. |
| pkg | Character vector. Bioconductor packages. |

## Author(s)

Ivan Jacob Agaloos Pesigan

PkgProjectGitHub *Install Project Packages from GitHub*

## Description

A package found in any library is never upgraded. It is installed when it is missing, or when its `ref` is a commit SHA that differs from the installed commit (`RemoteSha`). Dependencies are installed only when missing.

## Usage

```
PkgProjectGitHub(path, pkg, ref = "HEAD")
```

## Arguments

| | |
|---|---|
| path | Character string. Project path. |
| pkg | Character vector. GitHub packages. |
| ref | Character vector. Git branch, tag, or commit SHA (only a SHA reinstalls). |

## Author(s)

Ivan Jacob Agaloos Pesigan

PkgProjectVersion *Install Project Packages with Specific Version*

## Description

Install Project Packages with Specific Version

## Usage

```
PkgProjectVersion(path, pkg, ver)
```

## Arguments

| | |
|---|---|
| path | Character string. Project path. |
| pkg | Character vector. R packages. |
| ver | Character vector. Version. |

## Author(s)

Ivan Jacob Agaloos Pesigan

PkgRepo *Setup RStudio Package Manager Repository Based on the Current Operating System*

## Description

The repository is the Posit Package Manager snapshot of the image's date, so that a package the image lacks matches the image's other packages: the environment variable RPROJECT_SNAPSHOT (set in GitHub Actions from the lock), the date of the running sif, the lock's snapshot, or latest.

## Usage

```
PkgRepo(path)
```

## Arguments

| | |
|---|---|
| path | Character string. Project path. |

## Value

Returns the repository url.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Profile *Setup* .Rprofile

### Description

Setup .Rprofile

### Usage

```
Profile(path, project)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| project | Character string. Project name. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## Project *Setup the project.Rproj file*

### Description

Setup the project.Rproj file

### Usage

```
Project(path)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## Public *Check the Public Export*

### Description

Checks the working tree that `make auto` is about to publish (after `clone.sh` swaps in the public `.git`): no private files (e.g., `.setup/ai/`, `.setup/logs/`, `AGENTS.md`), no file larger than 50 MB, and no tokens (`SecretScan()`) in the files that would be published. `make auto` calls `Public()` before `git add`, so a failed check stops the push.

### Usage

```
Public(path)
```

### Arguments

| | |
|---|---|
| `path` | Character string. Project path. |

### Value

Returns the files that would be published, invisibly.

### Author(s)

Ivan Jacob Agaloos Pesigan

## Quarto *Render Quarto Project*

### Description

Render Quarto Project

### Usage

```
Quarto(path)
```

### Arguments

| | |
|---|---|
| `path` | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## ReadMe *Build README.md*

### Description

Build README.md

### Usage

```
ReadMe(path)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## SecretScan *Scan Project for Secrets*

### Description

Searches text files in the project for strings that look like tokens (GitHub, Docker Hub, Quarto Pub, Overleaf, Codecov, OpenAI, Anthropic). Matches are reported with the value masked.

### Usage

```
SecretScan(path, max_size = 5e+06, error = TRUE)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| max_size | Numeric. Files larger than max_size bytes are skipped. |
| error | Logical. If TRUE, stop with an error when secrets are found. |

### Value

Returns a data frame of findings with the columns file, line, type, and match (masked).

### Author(s)

Ivan Jacob Agaloos Pesigan

## Site *Build Website*

### Description

Build Website

### Usage

```
Site(path)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## SizeCheck *Check a Project Against Overleaf's Limits*

### Description

Measures what goes to Overleaf: every file in the project except local-only folders (`.git`, `.verify`, `.library`, `.Rproj.user`, `detritus`) and build output that `make` writes and Overleaf never gets (`docs`, `pkgdown`, `quarto`, `.setup/build`, `.setup/latex/pdf`, `.setup/quarto/bib`), the local-only data tiers (`.setup/data-external`, `.setup/data-cache`), and images (`.sif`).

### Usage

```
SizeCheck(
  path,
  editable = 7,
  text_file = 2,
  file = 50,
  total = 100,
  files = 2000L,
  error = FALSE
)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| editable | Numeric. Limit for all editable text, in MB. |
| text_file | Numeric. Limit for one editable text file, in MB. |
| file | Numeric. Limit for one file, in MB. |
| total | Numeric. Recommended limit for the whole project, in MB. |
| files | Integer. Limit for the number of files. |
| error | Logical. If TRUE, stop with an error when a check fails. |

### Details

- **Editable text** (fails): non-binary files in total above editable MB. Overleaf's GitHub sync stops above 7 MB.
- **Text file size** (fails): any non-binary file above text_file MB. Overleaf caps editable files at 2 MB.
- **File size** (fails): any file above file MB. Overleaf caps uploads at 50 MB.
- **Number of files** (fails): more than files files. Overleaf allows 2,000.
- **Total size** (warning): above total MB. Overleaf recommends under 100 MB with GitHub sync or git.
- **Duplicates** (warning): files of at least 1 MB with the same content.

The report also lists the largest text files and the largest files.

### Value

Returns a data frame of checks with the columns check, status, and detail.

### Author(s)

Ivan Jacob Agaloos Pesigan

---

## Spec *Create a Spec*

---

### Description

Creates .setup/ai/specs/YYYY-MM-DD-<topic>.md from a template. Every change set starts with a spec that the user approves.

### Usage

```
Spec(path, topic, title = topic)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| topic | Character string. Short topic in kebab case (e.g., "log-spec"). The change-set log uses the same topic. |
| title | Character string. Title of the spec. |

### Value

Returns the path of the spec invisibly.

### Author(s)

Ivan Jacob Agaloos Pesigan

---

## Style *Style Project*

---

### Description

Style Project

### Usage

```
Style(path)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |

### Author(s)

Ivan Jacob Agaloos Pesigan

---

## Tags *Library Tags*

---

### Description

Lists the key, annotation, and keywords of every entry in the project's Quarto bibliography (.setup/quarto/bib/quarto.bib, written by Bib()), for the "Library Tags" page of the Quarto site.

### Usage

```
Tags(path)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |

### Value

Returns a data frame with the columns key, annotation, and keywords (NA when an entry has none), sorted by key. Without the bibliography, an empty data frame, with a message.

### Author(s)

Ivan Jacob Agaloos Pesigan

---

## Template *Write the Template Files*

---

### Description

Writes the template files of the project's type from rProject (inst/template/), so that a template change reaches a project with make and no Overleaf refresh.

### Usage

```
Template(path, type = NULL, check = FALSE)
```

### Arguments

| | |
|---|---|
| path | Character string. Project path. |
| type | Character string. Project type. If NULL, it is read from .setup/scripts/project.R. |
| check | Logical. If TRUE, write nothing and report what would change. |

### Details

The type comes from a line in .setup/scripts/project.R, type <- "cran"; one of "rProject", "cran", "man", "lib", "talk", and "docker". Each type uses a few sets of files (inst/template/manifest.tsv). Each file has a tier:

- Tier 1, kept in Overleaf: the files Overleaf needs to compile the PDFs (the LaTeX preamble, .setup/latex/r-scripts/, .setup/load/load.R, style files, the lib documents).
- Tier 2, written by make: everything else (workflows, scripts, ...). These files are not in Overleaf or the private repository, but the Makefile (make needs it to run at all) and the .gitignore files (a fresh clone keeps its export rules), kept in Overleaf.

Every file is written when it is missing or differs (the template wins; change it in rProject). An overwritten tier 2 file is named; the files kept in Overleaf are named for upload. They are regular files in Overleaf, not links (if one still is, delete the link first); only .setup/r-dependencies/ link to the functions project. latexmkrc and project.Rproj, also regular files in Overleaf, are written by LatexCompile() and Project().

Besides the files of its sets, a project gets:

- The workflows it chooses from the catalog (`inst/template/workflows/`), named in `project.R`: for example `workflows <- c("docker-build-push-multi")`. The daily docker workflow (the one with `{{title}}`) is written as `<workflow>-<workflow_daily>.yml`, titled `workflow_title`. The daily builds run as a chain: an image with `workflow_cron` runs on that schedule; every image also starts when the image before it sends a `base-image-built` event; and an image with `workflow_next` (`"owner/repository"`, or several) sends that event to each next image when its build succeeds (with the `PAT` secret).
- The folder skeleton of its type: an empty `.gitignore` in each standard folder marked `create` in `inst/template/folders.tsv`, if the folder has none.

Files the template no longer has (`inst/template/removed.tsv`) are deleted and named, so that they can be deleted in Overleaf too. A folder on that list is deleted only when nothing but `.gitignore` files is left in it; otherwise it is named and kept. Nothing in a private folder is deleted.

In rProject itself, the tier 1 files are copied into `inst/template/` first, and the files are written from `inst/template/` in the project folder, not from the installed package.

Nothing is written or deleted in GitHub Actions (`GITHUB_ACTIONS` is `"true"`): CI uses the files already in the public repository.

## Value

Returns the paths of the files that were written (with `check = TRUE`, that would be written), invisibly, with the deleted files (or those that would be deleted) in the attribute `removed`.

## Author(s)

Ivan Jacob Agaloos Pesigan

---

`TinyTex` *Install* `~/.TinyTex`

---

## Description

Install `~/.TinyTex`

## Usage

```
TinyTex(path, bundle = "TinyTeX-2", force = FALSE)
```

## Arguments

| | |
|---|---|
| `path` | Character string. Project path. |
| `bundle` | The bundle name of TinyTeX (which determines the collection of LaTeX packages to install). See https://github.com/rstudio/tinytex-releases#releases for all possible bundles and their meanings. |
| `force` | Logical. Force install. |

### Author(s)

Ivan Jacob Agaloos Pesigan

---

## Tree *Project Folders and What They Are For*

---

### Description

Prints the standard folders of the project's type (`inst/template/folders.tsv`) with their purpose, and flags the folders that differ from the standard: a folder that `make` creates for the type but that is missing, and a folder that is not standard for the type.

### Usage

```
Tree(path)
```

### Arguments

| | |
|---|---|
| `path` | Character string. Project path. |

### Details

The type comes from the `type` line in `.setup/scripts/project.R` (see `Template()`); without it, it is guessed as in `AI()`.

### Value

Returns a data frame, invisibly, with the columns `folder`, `status` (`"present"`, `"missing"`, `"absent"` for an optional folder, `"not standard"` for this type, or `"unknown"`), `private`, and `purpose`.

### Author(s)

Ivan Jacob Agaloos Pesigan

---

## Verify *Verify an Overleaf Push*

---

### Description

Checks a push from Overleaf to the private GitHub repository against the change-set log in `.setup/logs`.

## Usage

```
Verify(
  path,
  private = file.path(path, ".verify"),
  log = NULL,
  record = FALSE,
  error = TRUE
)
```

## Arguments

| | |
|---|---|
| path | Character string. Project path. |
| private | Character string. Path to a clone of the private repository at the pushed commit. By default .verify in the project. |
| log | Character string. File name of the change-set log in .setup/logs. If NULL, the log named in the pushed commit message (Log: .setup/logs/<log>), else the newest log with a pending push in .setup/logs/index.md. |
| record | Logical. If TRUE, write the commit and the result in the log's "Pushes" table and in .setup/logs/index.md, filling the pending row, or adding a row when there is none. |
| error | Logical. If TRUE, stop with an error when a check fails. |

## Details

The checks, and when each fails:

- **New commit:** the commit is already recorded in index.md (.verify is stale, or there was no new push). Nothing is recorded.
- **Listed files:** a file listed under "Files to update in Overleaf" is missing from or differs in the private repository. The list is the one of the pushed commit: the section of the log with a code block that starts with the commit's first line (the top-level lists for the log's own commit message; the lists inside a later push's ## section, none for a push of records only). If no block matches, every list in the log is used.
- **Deletions:** a file listed under "Delete in Overleaf" is still in the private repository.
- **Unlisted changes:** (warning only) other files differ between the project and the private repository (.setup/notes/claude.txt, the resume commands, and .setup/todo/, the user's to-do list, are not counted).
- **Commit message:** (warning only) the last commit message does not end with Log: .setup/logs/<log>.
- **Secrets:** SecretScan() finds a token in the private repository.
- **Public export:** a file that make auto would publish is private (e.g., .setup/ai/, .setup/logs/, .setup/repo/, AGENTS.md) or larger than 50 MB.
- **Repo info:** .setup/repo/private-overleaf.md is not an Overleaf project URL (for example, a share link, which grants access). Warning only: a file in .setup/repo/ is missing, private-github.md differs from the .verify remote, or public-github.md differs from the repository in clone.sh.

Generated files (e.g., NAMESPACE, man/), local folders (e.g., .library), and secret files are never compared.

The private repository must be cloned on the host, because SSH keys are not available in the container. .setup/scripts/verify.sh clones or pulls it into .verify (local only; never uploaded) and then calls Verify().

## Value

Returns a data frame of checks with the columns check, status, and detail.

## Author(s)

Ivan Jacob Agaloos Pesigan

---

# VignettesPrecompile *Precompile Vignettes*

---

## Description

Precompile Vignettes

## Usage

```
VignettesPrecompile(path)
```

## Arguments

| | |
|---|---|
| path | Character string. Project path. |

## Author(s)

Ivan Jacob Agaloos Pesigan

---

# z *Data from the Standard Normal Distribution*

---

## Description

Data from the Standard Normal Distribution

## Usage

```
z
```

## Format

A vector of length 100.

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